

Drone Dynamic Testing After Midterm - Reliability Notes

This English version is formatted for graduate application review. It preserves the engineering purpose, design decisions, validation evidence, and individual contribution signals of the original course document while presenting the material in concise English.

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Project	Quadcopter Ball-Transfer Drone
Course	Practice of Mechanical Engineering
Document Type	System test record
Primary Role	Team lead and flight-test debugger
Original Source	drone-dynamic-testing-after-midterm.pdf (2 pages)

Executive Summary

This record summarizes dynamic testing after the midterm stage, when the project moved from basic assembly toward repeated flight behavior and mission execution.

The major value of the work was diagnostic: drift, unstable response, crash risk, frame damage, and ball-transfer issues had to be separated into controllable causes rather than discussed as vague flight failure.

Applicant Contribution Signals

- Converted test failures into subsystem-level hypotheses.
- Connected flight response to RC setup, deadband, control parameters, structure, and payload behavior.
- Used repeated testing to improve final reliability rather than waiting for a perfect build.

Observed Problems

Dynamic flight introduced problems that were not visible during static build checks, including drift, unstable attitude response, transfer instability, and crash-related damage.

These symptoms were treated as engineering information, not simply as failed attempts.

Debugging Method

The team separated control input behavior, flight-controller settings, frame stiffness, payload placement, and operator procedure.

Each test created a clearer next action: recalibrate, reinforce, revise geometry, change procedure, or reduce risk during final attempts.

Technical Decision Table

Constraint	Decision	Why It Mattered	Skill Demonstrated
Drift and instability	Checked control setup and neutral behavior	Reduced hidden bias in flight commands	Controls debugging
Crash damage	Reviewed structure and repair plan	Kept the system testable after failures	Hardware iteration
Transfer reliability	Adjusted procedure and payload interaction	Protected mission success, not just flight	System-level testing

How to Read This Evidence

This document is intended to help a reviewer quickly understand the engineering content of the original report in English. The original PDF remains available in the project archive as the source artifact with its original figures, tables, screenshots, and course formatting.